

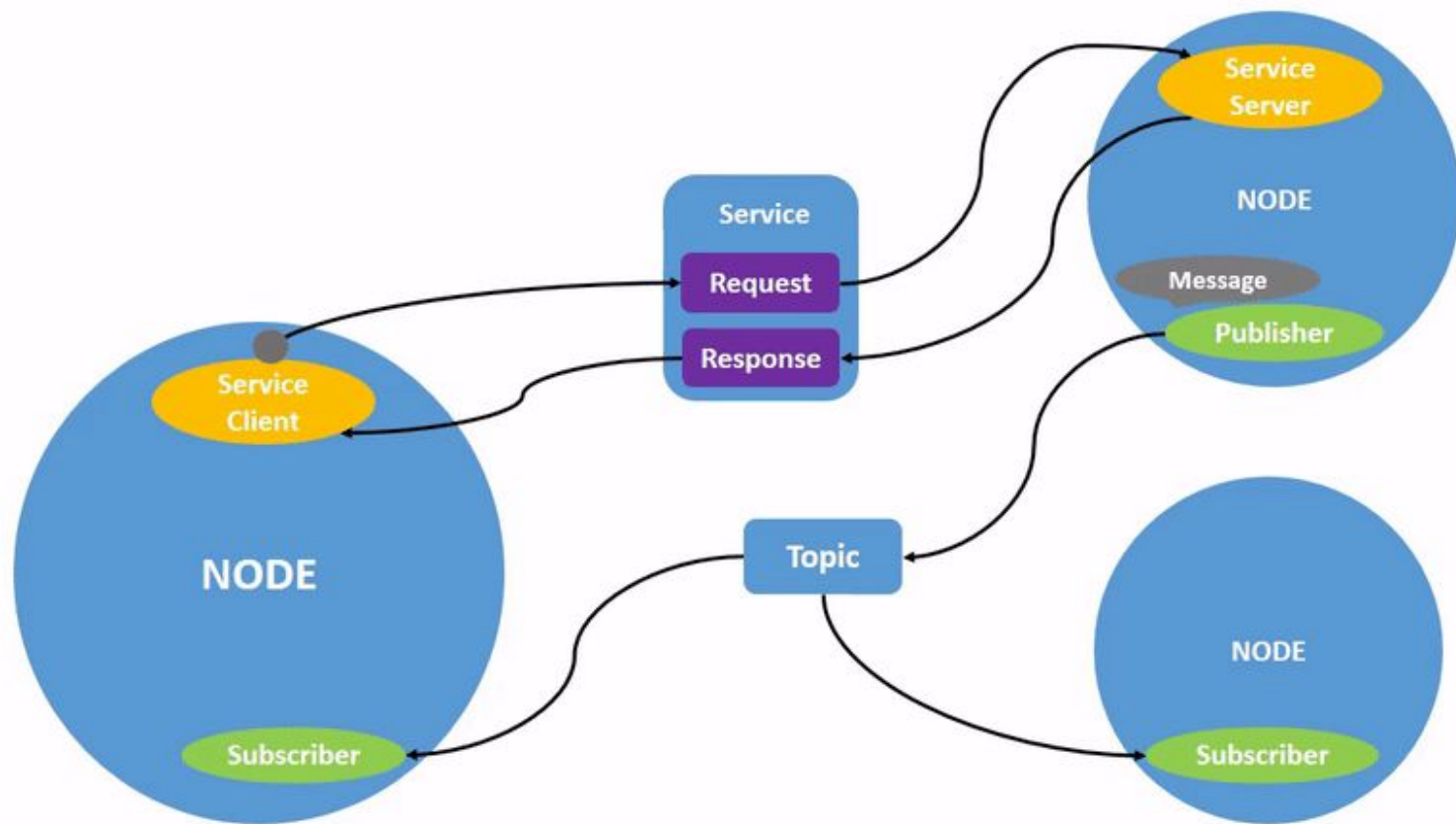
CISC 667

Introduction to Autonomous Driving

Lecture 1
Sept 3, 2025

Today's Topics

- ROS Basics



Common ROS2 Commands

- List all available topics
 - `ros2 topic list`
- Echo a topic's content
 - `ros2 topic echo <topic_name>`
- Check a topic's type/publisher/subscriber/QoS
 - `ros2 topic info -v <topic_name>`
- Bring up the visualization tool
 - `rviz2`
- List all available nodes
 - `ros2 node list`
- Check a node's current subscriber/publisher/services
 - `ros2 node info <node_name>`
- Record data
 - `ros2 bag record -o <bag_name>`
- Play data
 - `ros2 bag play <bag_name>`

Common ROS2 Commands

- Source the executables from ROS
 - `source /opt/ros/humble/setup.bash`
- Source the executables from you custom ROS packages
 - `source path_to_package/install/setup.bash`
- Run a ROS2 executable
 - `ros2 run <package_name> <script_name>`
- Launch a ROS2 executable
 - `ros2 launch <package_name> <launch_file_name>`
- Compile Package
 - `cd path_to_package`
 - `colcon build --symlink-install`

Creating a Package

- Create your workspace
 - `$HOME/group_name/member_name/src`
- Create your package
 - `ros2 pkg create --build-type ament_cmake --license Apache-2.0 <pkg-name> --dependencies [deps]`
 - `ros2 pkg create --build-type ament_python --license Apache-2.0 <pkg-name> --dependencies [deps]`
- <https://docs.ros.org/en/humble/How-To-Guides/Developing-a-ROS-2-Package.html>

List of Challenges

1. Visualize both the camera and LiDAR topics in RVIZ.
2. What is the average publishing frequency of the camera and LiDAR topics?
3. What is the dimension of the camera images in the camera topic?
4. How many points are contained in each LiDAR message?
5. What is the average size of each message of camera and LiDAR topics?
 - a. how is it calculated?
6. Write a script that is the bandpass filter returning only LiDAR points with distance (radius) between 10 and 50 meters.